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Theory: Dominance effects

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Bayer CropScience

AlphaSimR Implementation of Dominance

- Dominance effects are a function of additive effects

$$d = \delta|a|$$

- User specifies distribution of dominance degrees

$$\delta \sim N(\mu_\delta, \sigma_\delta^2)$$

- No dominance: $\delta = 0$
- Complete dominance: $\delta = 1$
- Partial dominance: $0 < \delta < 1$
- Over-dominance: $\delta > 1$

Average Effect of an Allele Substitution

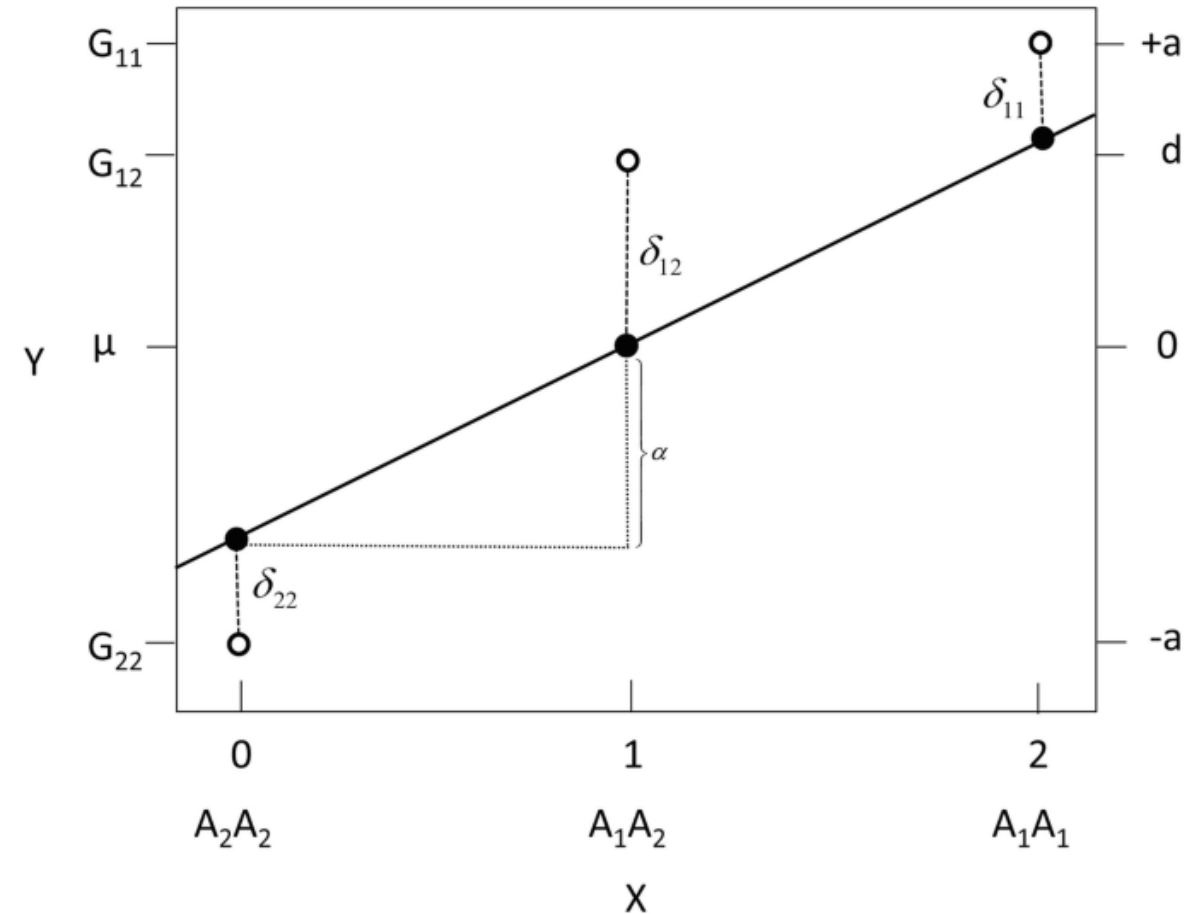
- The textbook formula assumes HWE

$$\alpha = a + (q - p)d$$

- General formula not assuming HWE

$$\alpha = a + \frac{1-F}{1+F} (q - p)d$$

- It is just a regression coefficient
 - Regress genetic value on genotype dosage
 - Assuming a base population without LD
 - Many other descriptions require HWE
 - See Falconer 1985
 - Allows for generalizations to other cases



Breeding Values

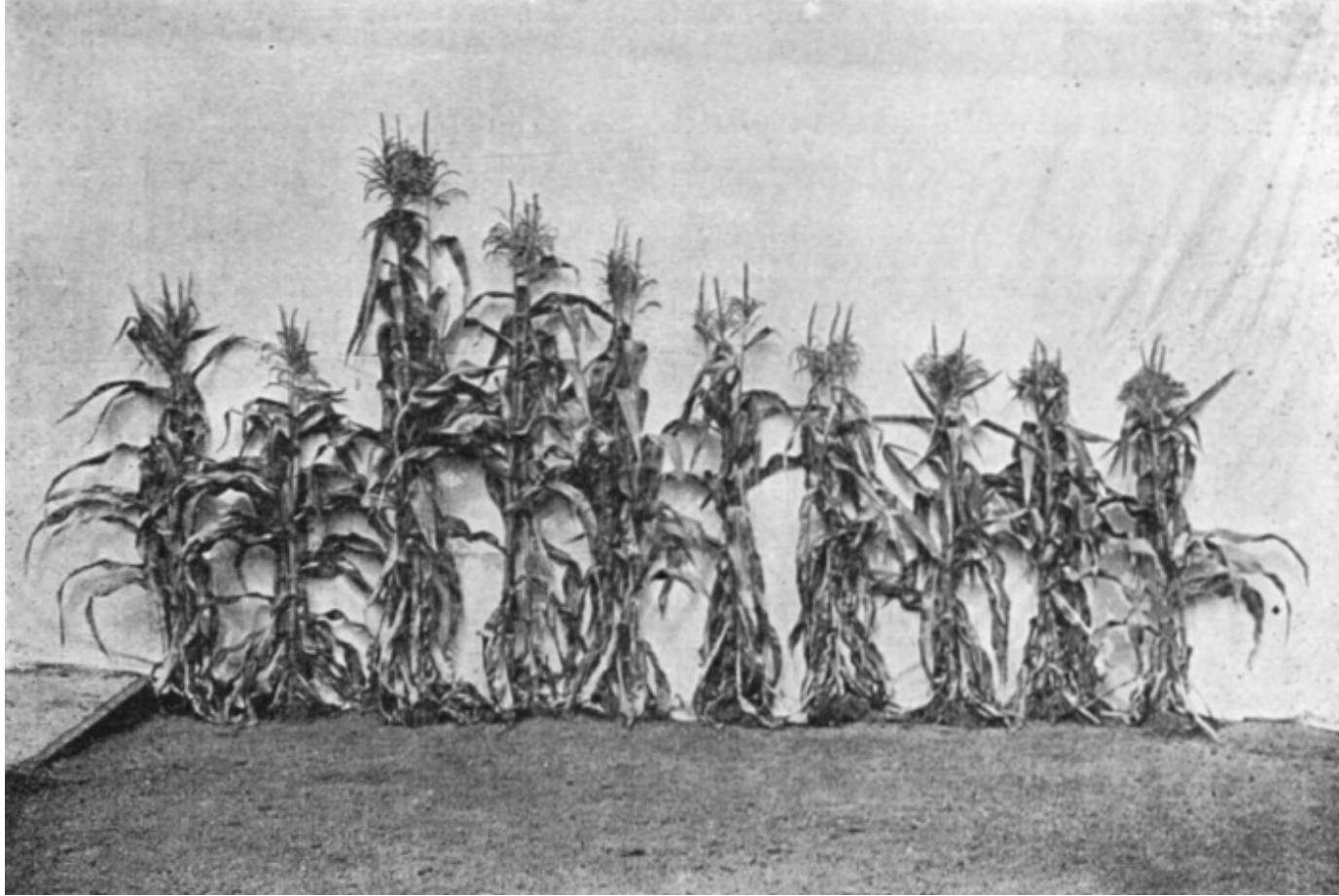
- Average effects are used to calculate breeding values
 - Breeding values are points on the regression line
- Breeding values used to calculate additive genetic variance
 - Remember this is not variance of additive effects
- There are some special properties of breeding values
 - Only valid under HWE and random mating
 - See Falconer 1985

Dominance Deviations

- Used to calculate dominance variance
- You can think of these as the lack-of-fit in a regression
 - Breeding values came from a regression
- You can also think of them as a quadratic polynomial
 - Constructed to be orthogonal
 - This thinking is needed for polyploids

AlphaSimR Demonstration

Heterosis and Inbreeding Depression

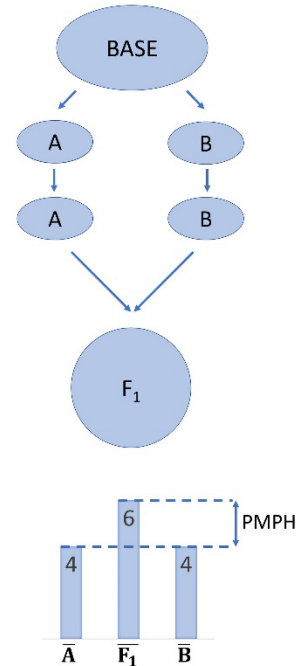


Source: Jones (1924)

Quantitative Genetics of Heterosis

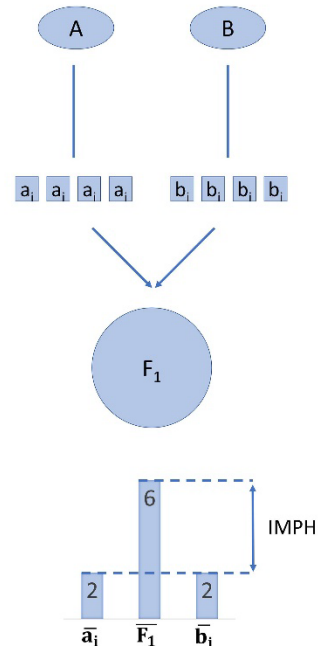
Crossbreeding,
Animals

Panmictic-midparent
heterosis



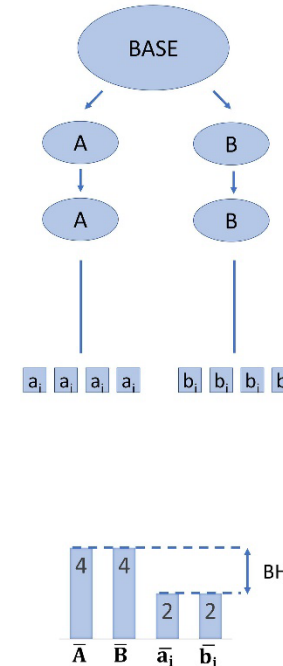
Hybrids,
Plants

Inbred-midparent
heterosis



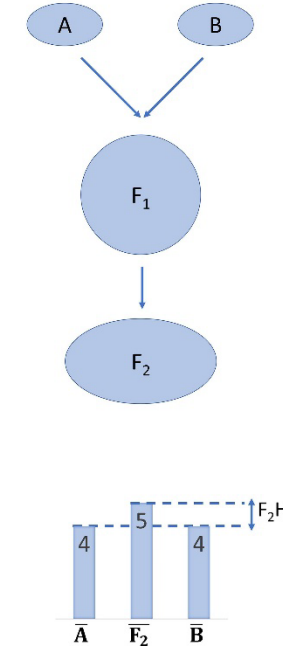
Inbred vs
Outbred

Baseline
heterosis



Inbreeding
Across Lines

F₂ heterosis



$$\begin{aligned}\bar{F}_1 &= (p - q - y)a + [2pq + y(p - q)]d \\ \bar{F}_2 &= (p - q - y)a + [2pq + y(p - q) - \frac{1}{2}y^2]d \\ PMPV &= (p - q - y)a + [2pq + y(p - q) - y^2]d \\ IMPV &= (p - q - y)a\end{aligned}$$

$$\begin{aligned}PMPH &= y^2d \\ F_2H &= \frac{1}{2}y^2d \\ BH &= 2\bar{p}\bar{q}d - \frac{1}{2}y^2d \\ IMPH &= 2\bar{p}\bar{q}d + \frac{1}{2}y^2d\end{aligned}$$

Source: Labroo *et al.* (2021)

Maize Breeding Program

- Want to model genomic selection (GS) in a maize program
 - Produces hybrids and selects on general combining ability (GCA)
 - Expect GS accuracy to be lower when adding dominance
 - GCA represents a moving target
- What we know
 - Dominance gene action can explain heterosis
 - Evidence for strong partial dominance in maize
 - Measure of long-term genetic gain for inbreds and hybrids

Dominance Theories for Heterosis

Genotype	0	1	2
Genetic Value	-a	d	a

- Overdominance
 - $d > a$
- (Partial) Dominance
 - $a > d > 0$
 - Pseudo-overdominance

Pseudo-overdominance Example

Locus 2 (a=1, d=0.1)	Locus 1 (a=1, d=0.1)		
	0	1	2
0	-2a (-2)	d-a (-0.9)	0 (0)
1	d-a (-0.9)	2d (0.2)	a+d (1.1)
2	0 (0)	a+d (1.1)	2a (2)

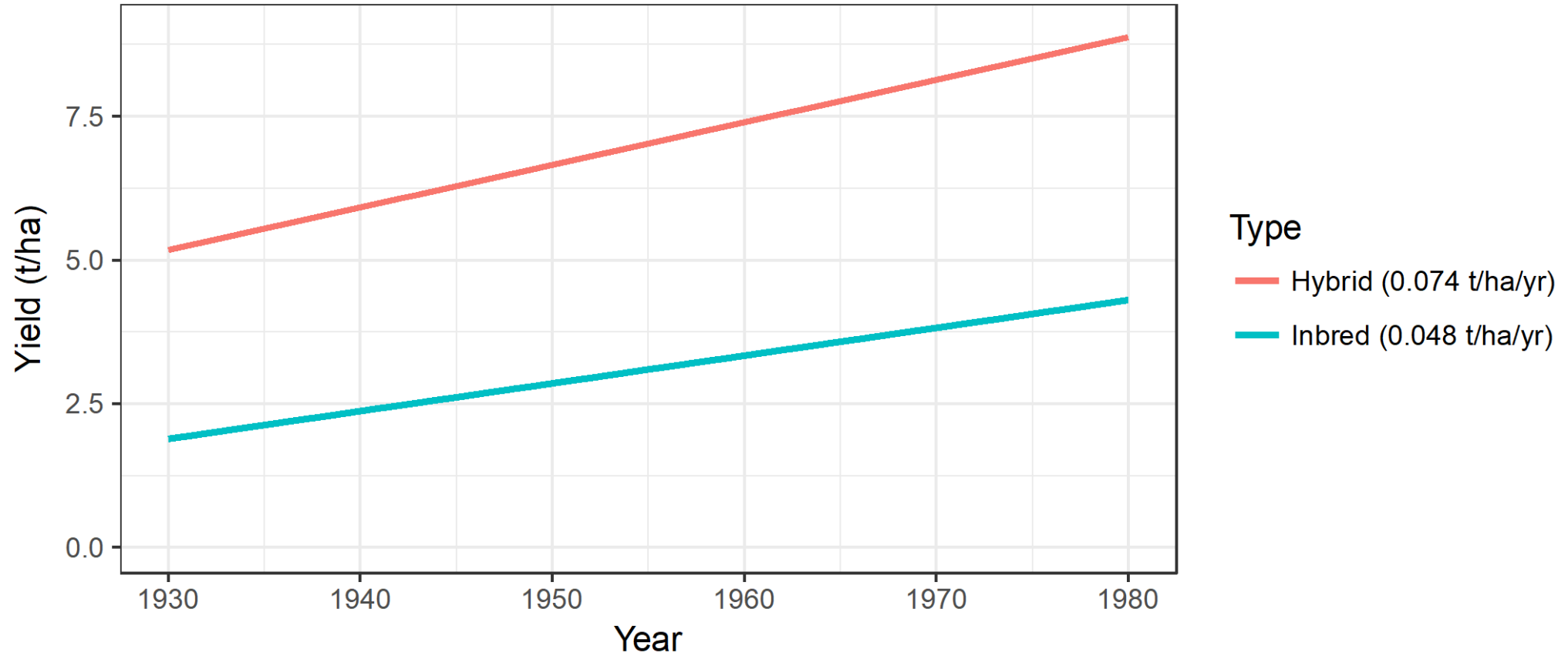
- All possible combinations
- Homozygote is best

Pseudo-overdominance Example

Locus 2 (a=1, d=0.1)	Locus 1 (a=1, d=0.1)		
	0	1	2
0	-2a (-2)	d-a (-0.9)	0 (0)
1	d-a (-0.9)	2d (0.2)	a+d (1.1)
2	0 (0)	a+d (1.1)	2a (2)

- Repulsion linkage (complete)
 - Unobserved combinations
- Heterozygote is best

Long-term Genetic Gain in Maize



Source: Troyer and Wellin (2009)

Quantitative Genetics for Heterosis (Hybrids)

- Inbred midparent heterosis
 - Deviation of hybrids from midparent value
 - Heterosis from two sources
 - Recovery of inbreeding
 - Split between pools

$$H_{IMP} = \sum_{i=1}^{n_{QTL}} 2\bar{p}_i\bar{q}_id_i + \frac{1}{2}(p_{1i} - p_{2i})^2d_i$$

Quantitative Genetics in AlphaSimR

$$a \sim N(0, \sigma_a^2)$$

$$d = \delta |a|$$

$$\delta \sim N(\mu_\delta, \sigma_\delta^2)$$

$$E(H_{IMP}) = \frac{\mu_\delta \sigma_G \sum_{i=1}^{n_{QTL}} 2 \bar{p}_i \bar{q}_i + \frac{1}{2} (p_{1i} - p_{2i})^2}{\sqrt{2 \pi \sum_{i=1}^{n_{QTL}} \bar{p}_i \bar{q}_i}}$$

Quantitative Genetics in AlphaSimR (cont'd)

$$E(H_{IMP}) = \frac{\mu_{\delta} \sigma_G \sum_{i=1}^{n_{QTL}} 2\bar{p}_i \bar{q}_i + \frac{1}{2} (p_{1i} - p_{2i})^2}{\sqrt{2\pi \sum_{i=1}^{n_{QTL}} \bar{p}_i \bar{q}_i}}$$

$$H_{IMP}(\sigma_G) \propto \sigma_G$$

$$H_{IMP}(\delta) \propto \mu_{\delta}$$

$$H_{IMP}(n_{QTL}) \propto y$$

$$y \approx \sqrt{n_{QTL}}$$

Setting Dominance Level for Maize

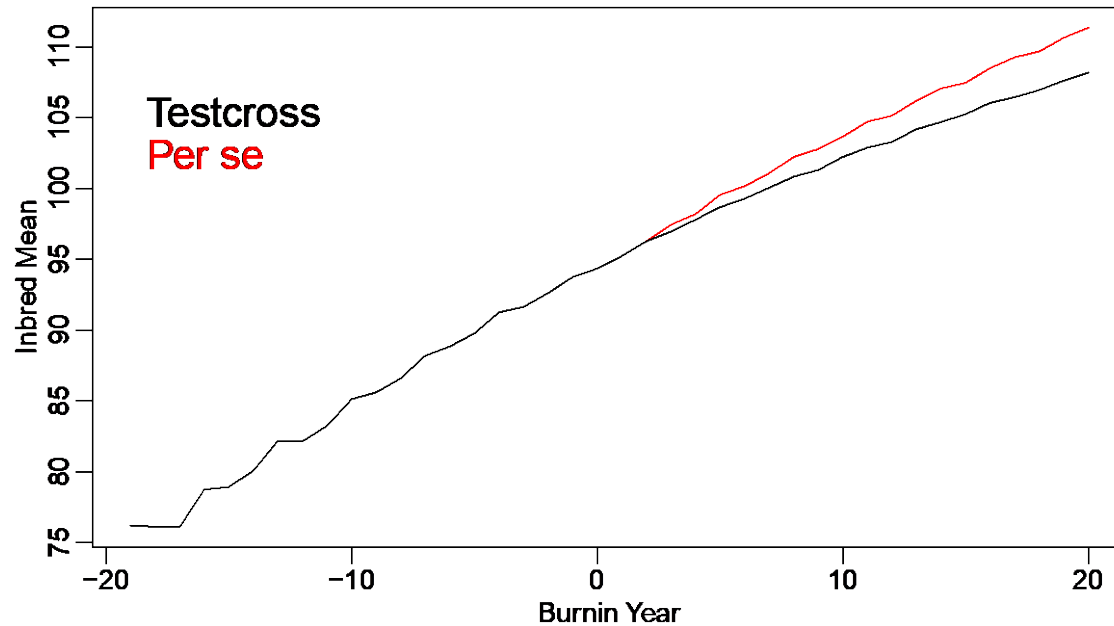
- Trade-off between two key values
 - Number of QTL
 - Mean dominance degree
 - Others matter to a lesser degree
- Estimated optimal values
 - 300 QTL per chromosome
 - 0.92 mean dominance degree
- Provides a good approximation to long-term genetic gain
 - Reasonable but not necessarily realistic

About that Troyer and Wellin Paper

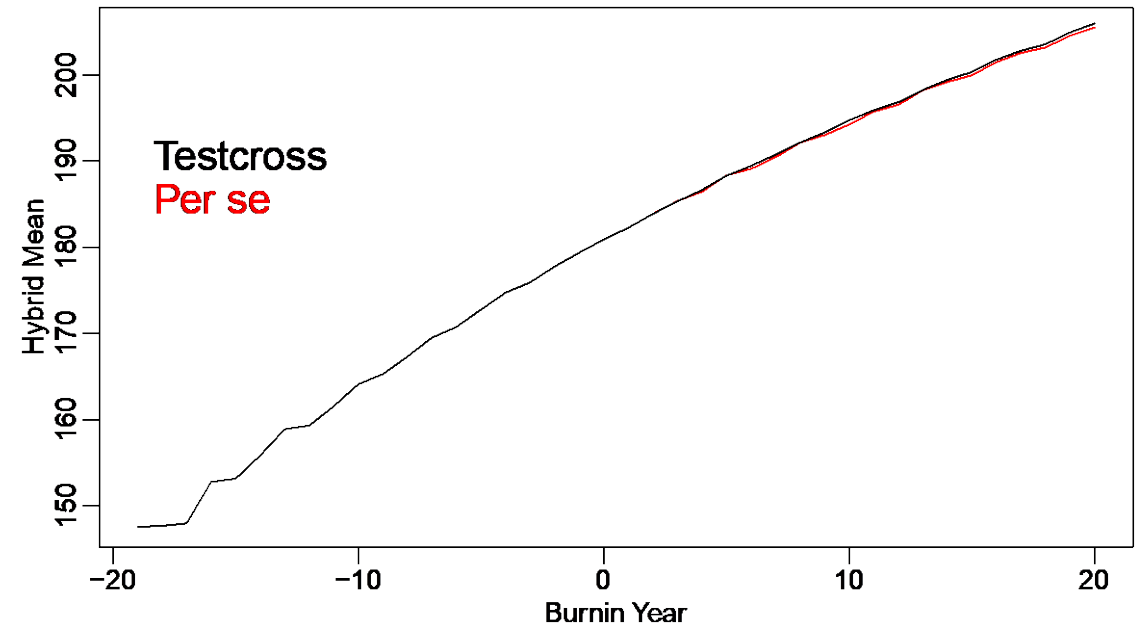
- They argued for eliminating the first round of testcrosses
 - To be replaced with a *per se* evaluation of inbreds
- We can examine this in simulation
 - Using the dominance values from before
- Simulation supports their recommendation

Replacing First Testcross with *per se* Evaluation

Inbred genetic gain



Hybrid genetic gain



AlphaSimR Demonstration

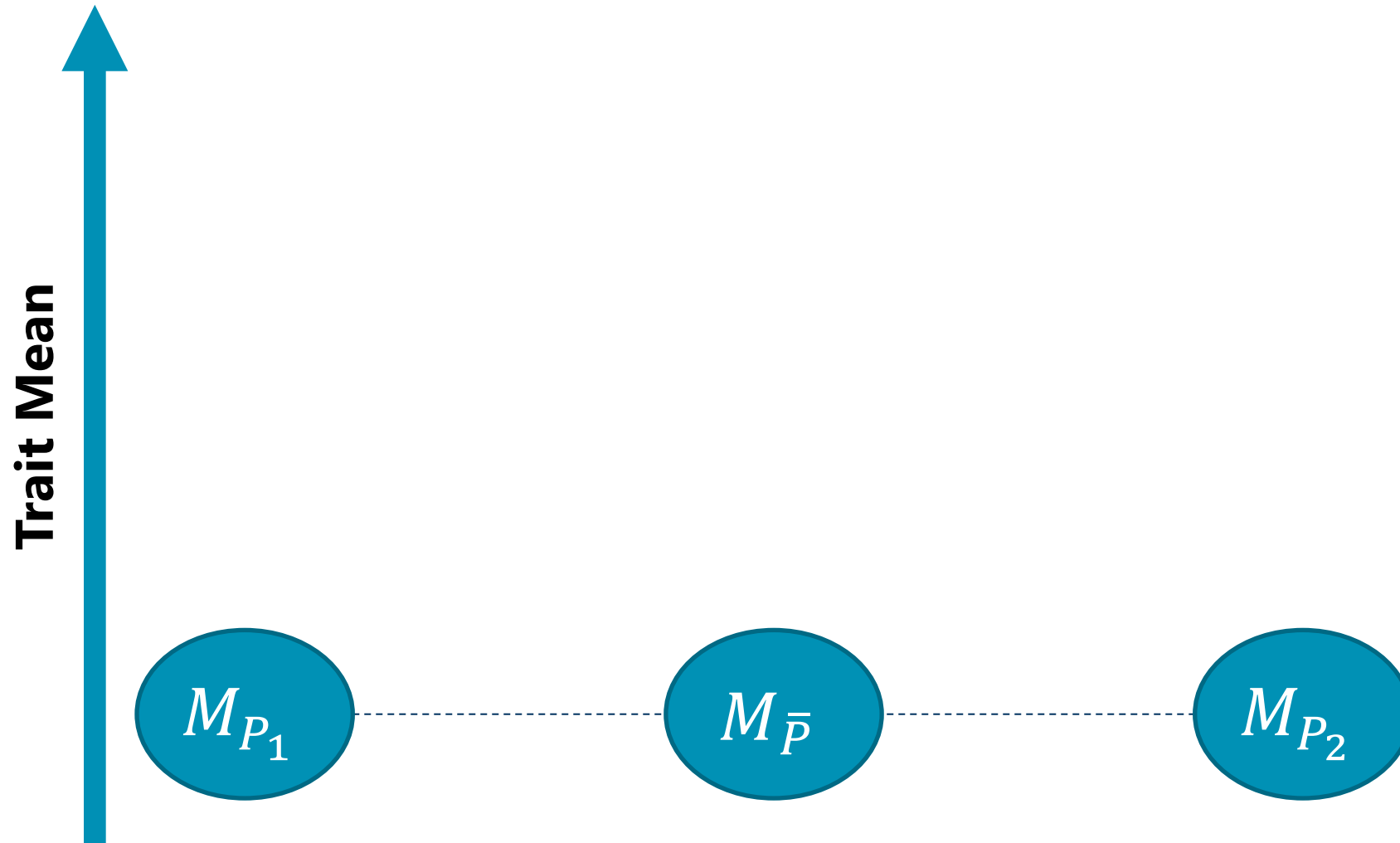
Heterosis in Animal Breeding

- Exploited with crossbreeding
 - Terminal crossbreeding similar to maize
- Results aren't as drastic as in maize
 - Animals aren't fully inbred
 - Lines/breeds are somewhat inbred
 - Exploiting panmictic midparent heterosis

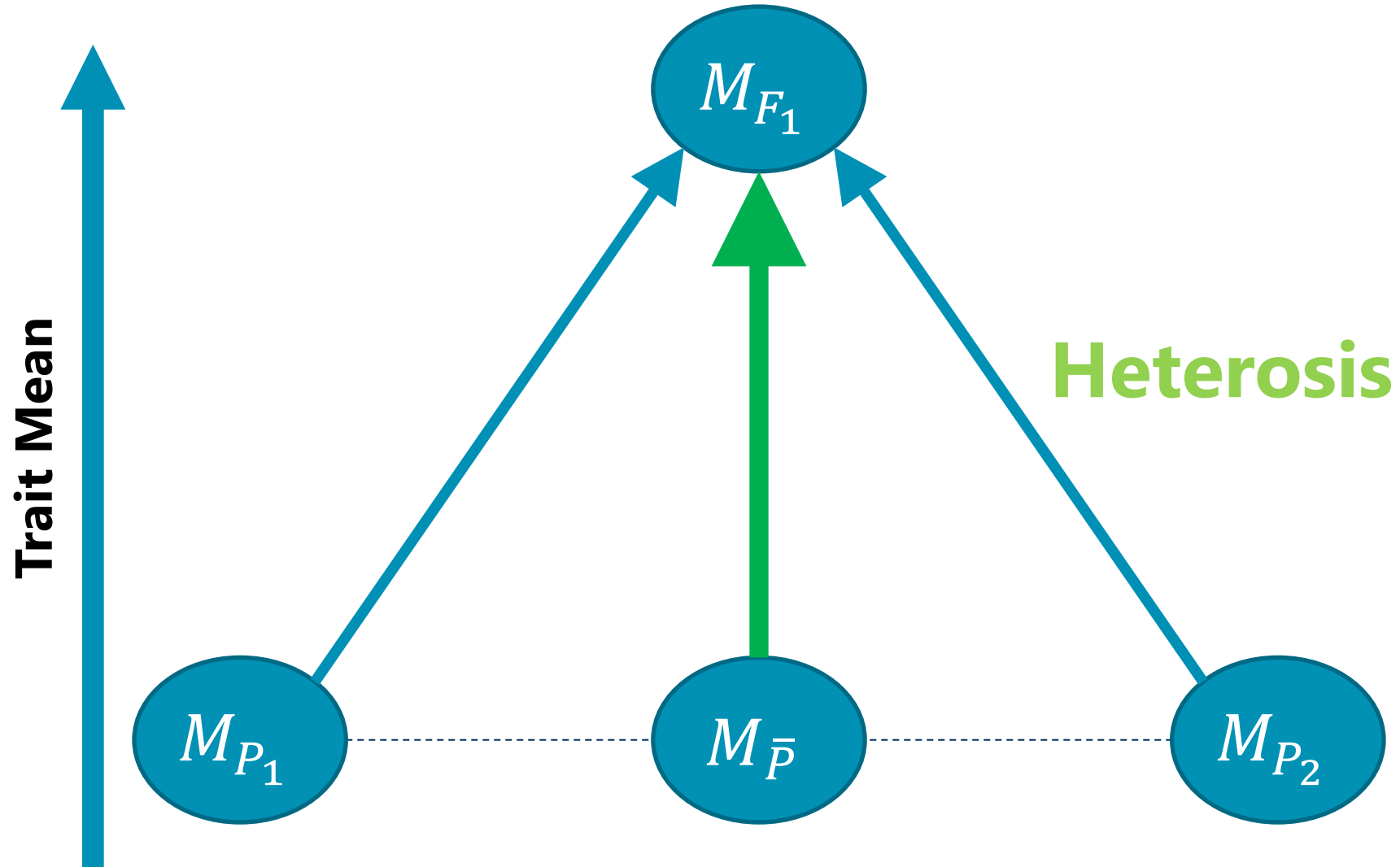
Two Breeds or Lines (HWE)



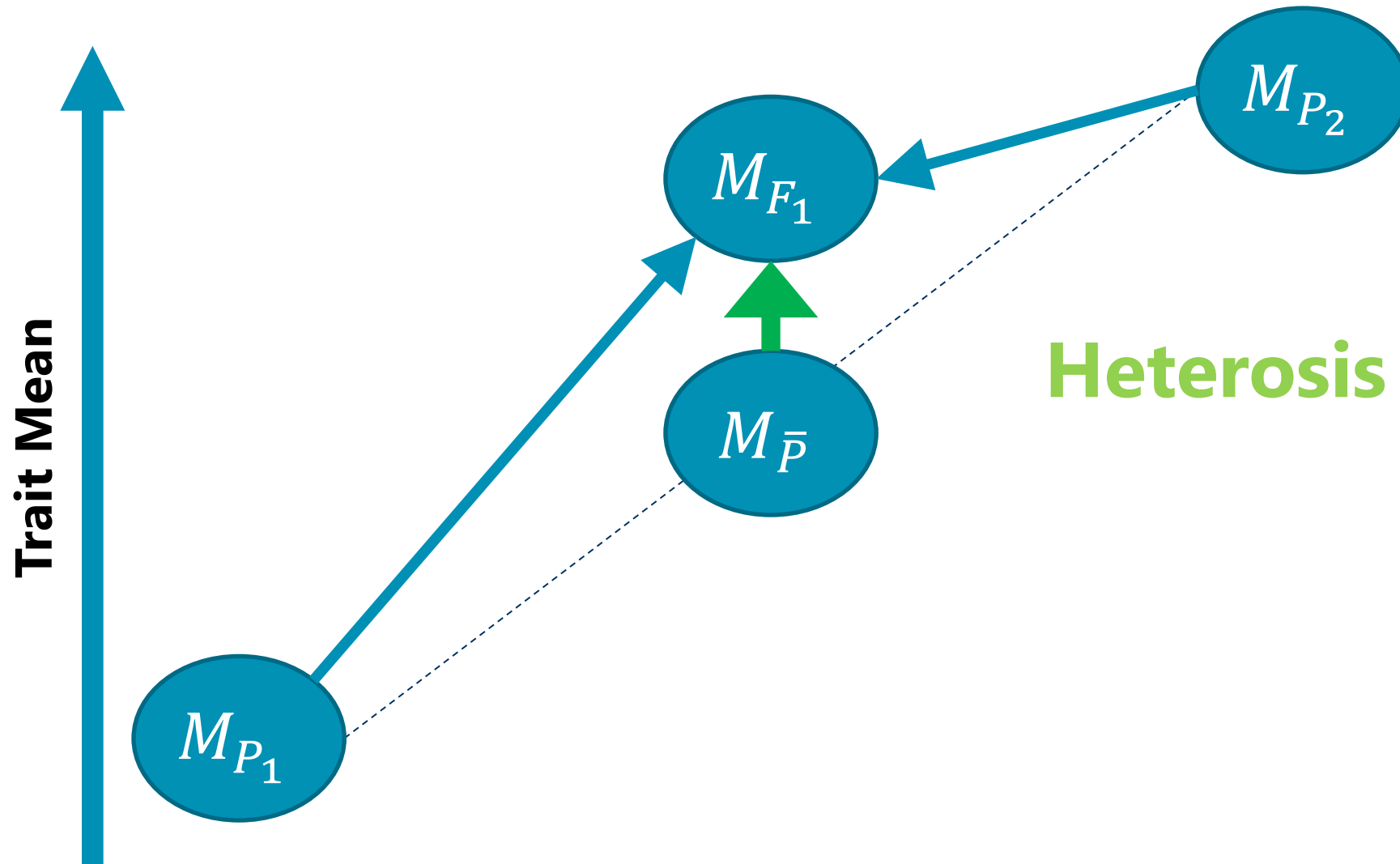
Midparent Value



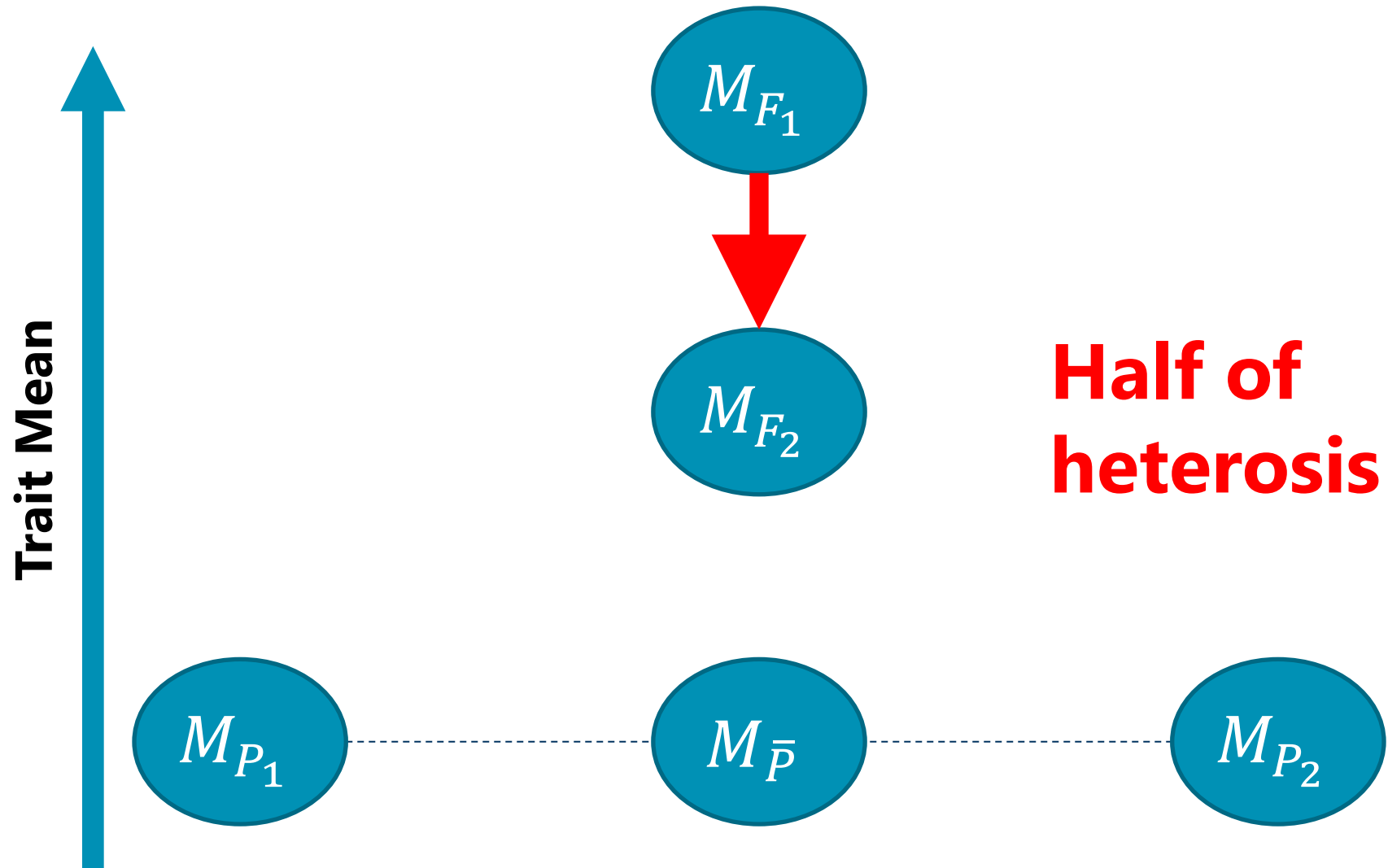
Crossbred Animals



Unequal Means



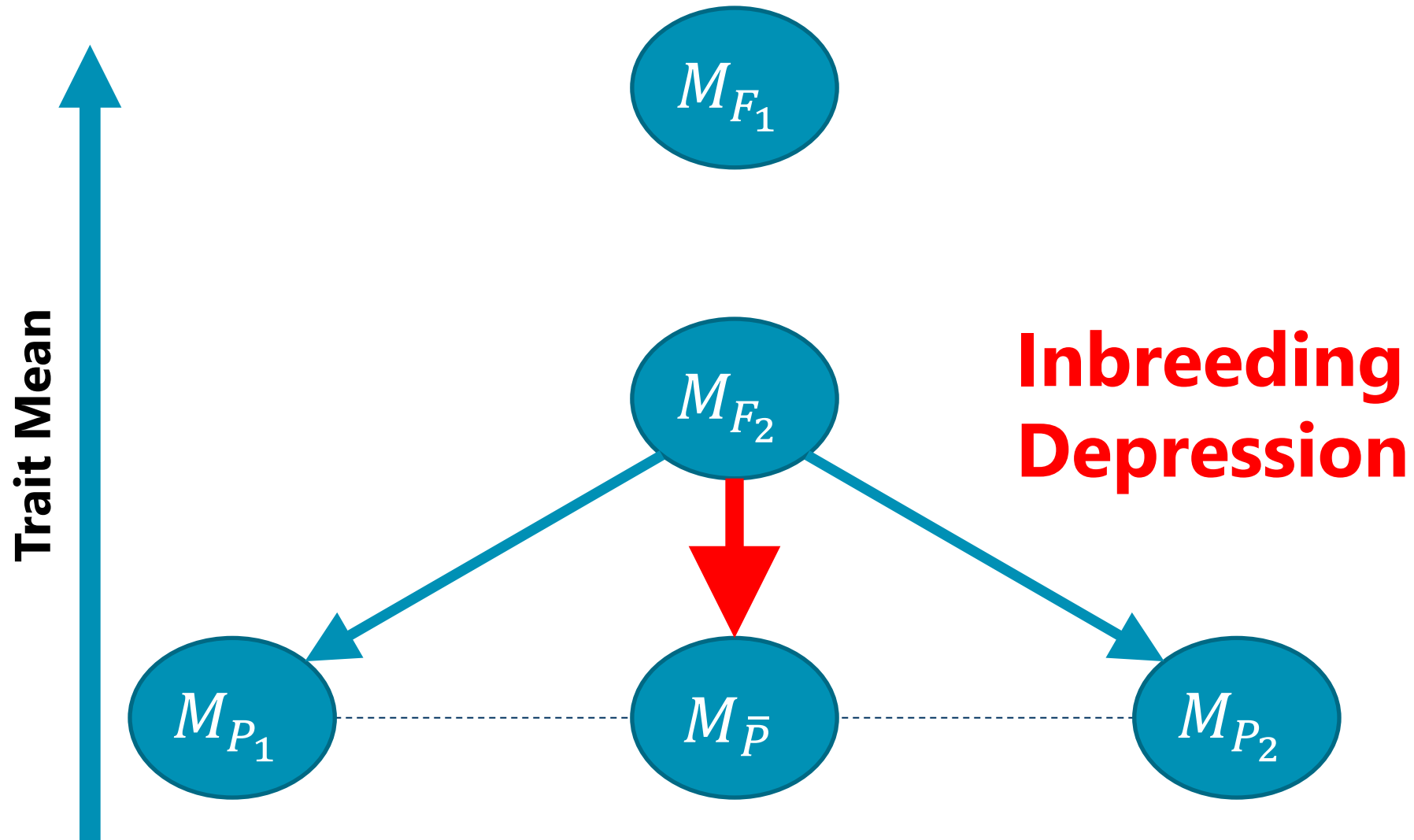
Intermate Crossbreds (HWE)



Inbreeding Depression in Animals

- Important for all breeding programs
 - Avoid mating relatives
- Textbook example of inbreeding
 - Split population into lines
 - The lines become inbred
- **Key point:** inbreeding is a relative value

Line Breeding (without Selection)



Inbreeding Depression

Population mean (no epistasis)

$$M_F = \sum (p - q)a + 2pq(1 - F)d$$

Decrease in mean due to inbreeding

$$I = M_{F=0} - M_{F=1} = \sum 2pqd$$

Heterosis

Population means

$$M_{P_1} = \sum (p - q)a + 2pqd$$

$$M_{P_2} = \sum (p' - q')a + 2p'q'd$$

Panmictic Midparent Heterosis

$$H = M_{F_1} - M_{\bar{P}} = \sum (p - p')^2 d = \sum 4\bar{p}\bar{q}Fd$$

Simulating QTL effects

Additive effects

$$a \sim N(0, \sigma_a^2)$$

Dominance effects

$$d = \delta |a|$$

Dominance degrees

$$\delta \sim N(\mu_\delta, \sigma_\delta^2)$$

Mean of dominance effects

$$E(d) = E(\delta)E(|a|) = \mu_\delta \sigma_a \sqrt{2/\pi}$$

Inbreeding Depression in Simulation

$$E(I) = \sum 2pqE(d) = \mu_{\delta}\sigma_a\sqrt{2/\pi} \sum 2pq$$

- Depends on user supplied parameters
 - Mean dominance degree
 - Number of QTL
- Depends on additive effect variance

Heterosis in Simulation

$$E(H) = \sum (p - p')^2 E(d) = \mu_{\delta} \sigma_a \sqrt{2/\pi} \sum (p - p')^2$$

- Depends on user supplied parameters
 - Mean dominance degree
 - Number of QTL
 - Allele frequency difference
- Depends on additive effect variance

Additive Effect Variance (σ_a^2)

- Tuned for additive genetic variance (V_A)
 - User specified value
- Accomplished using linear scaling
 1. Sample effects from a standard normal
 2. Calculate variance
 3. Calculate and apply scaling constant

Behind the Scenes

Variance scaled using a single breed (HWE)

Additive genetic variance

$$V_A = 2 \sum pq \alpha^2 + 4 \sum_{i < j} D_{i,j} \alpha_i \alpha_j$$

Average effect

$$\alpha = a + d(q - p)$$

Average Effect Expectations

$$\begin{aligned} E(\alpha^2) &= E(a^2) + E(d^2)(q - p)^2 + 2E(ad)(q - p) \\ &= \sigma_a^2 \left[1 + (\mu_\delta^2 + \sigma_\delta^2)(q - p)^2 \right] \end{aligned}$$

$$\begin{aligned} E(\alpha_1 \alpha_2) &= E(a_1)E(a_2) + \cdots + E(d_1)E(d_2)(q_1 - p_1)(q_2 - p_2) \\ &= \sigma_a^2 \mu_\delta^2 \left(\frac{2}{\pi} \right) (q_1 - p_1)(q_2 - p_2) \end{aligned}$$

Expectation for σ_a^2

$$E(\sigma_a^2) = \frac{V_A}{C_1 + (\mu_\delta^2 + \sigma_\delta^2)C_2 + \mu_\delta^2 C_3}$$

$$C_1 = 2\Sigma pq$$

$$C_2 = 2\Sigma pq(q - p)^2$$

$$C_3 = \frac{8}{\pi} \sum_{i < j} (q_i - p_i)(q_j - p_j) D_{i,j}$$

Expectation for Inbreeding Depression

$$E(I) = 2\Sigma pqE(d) = \mu_{\delta}\sigma_a C_1 \sqrt{2/\pi}$$

$$E(I) = \frac{\mu_{\delta} C_1 \sqrt{2V_A/\pi}}{\sqrt{C_1 + (\mu_{\delta}^2 + \sigma_{\delta}^2) C_2 + \mu_{\delta}^2 C_3}}$$

- Expectation for heterosis is similar

Deciphering the Equation

$$E(I) = \frac{\mu_{\delta} C_1 \sqrt{2V_A/\pi}}{\sqrt{C_1 + (\mu_{\delta}^2 + \sigma_{\delta}^2) C_2 + \mu_{\delta}^2 C_3}}$$

- Increasing V_A increases inbreeding depression
 - Use $\frac{I}{\sqrt{V_A}}$ for making comparisons
- Increasing σ_{δ}^2 decreases inbreeding depression

Simplifying Assuming $p = 0.5$

$$E(I) = \mu_{\delta} \sqrt{\frac{V_A n_{QTL}}{\pi}}$$

$$E(I) \propto \mu_{\delta}$$

$$E(I) \propto \sqrt{n_{QTL}}$$

Key point: trade-off between μ_{δ} and n_{QTL}

Practical Application

- No unique solution
- Narrow search space if possible
 - Dominance variance
 - Purebred-crossbred correlations
- Test a range of values
 - Checks sensitive to assumptions
 - Use the equations to help with this

Concluding Comments

- Incorporating dominance gets tricky
- Make sure doing so is necessary
 - Strategies for avoiding inbreeding depression
 - Strategies for exploiting heterosis
- Remember all models are wrong
 - We are seeking a useful model

AlphaSimR Demonstration